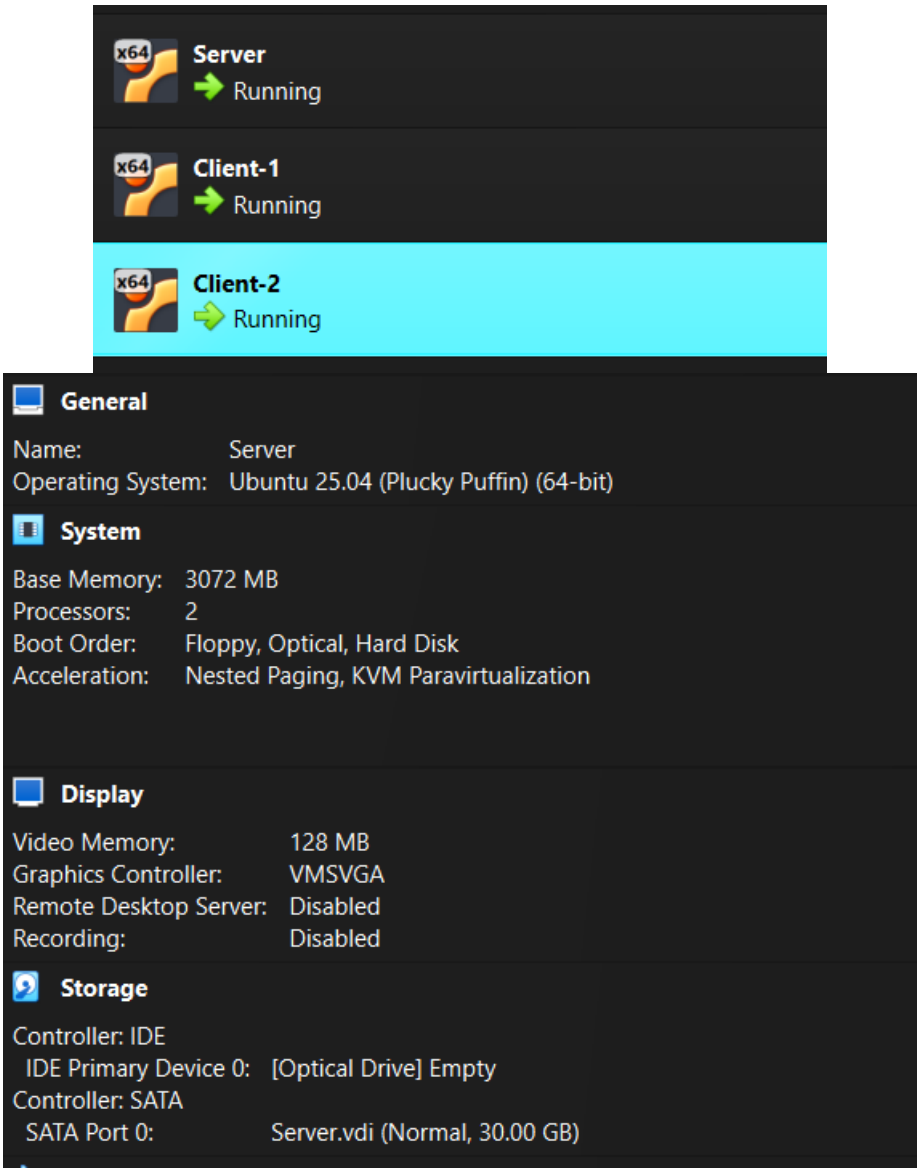
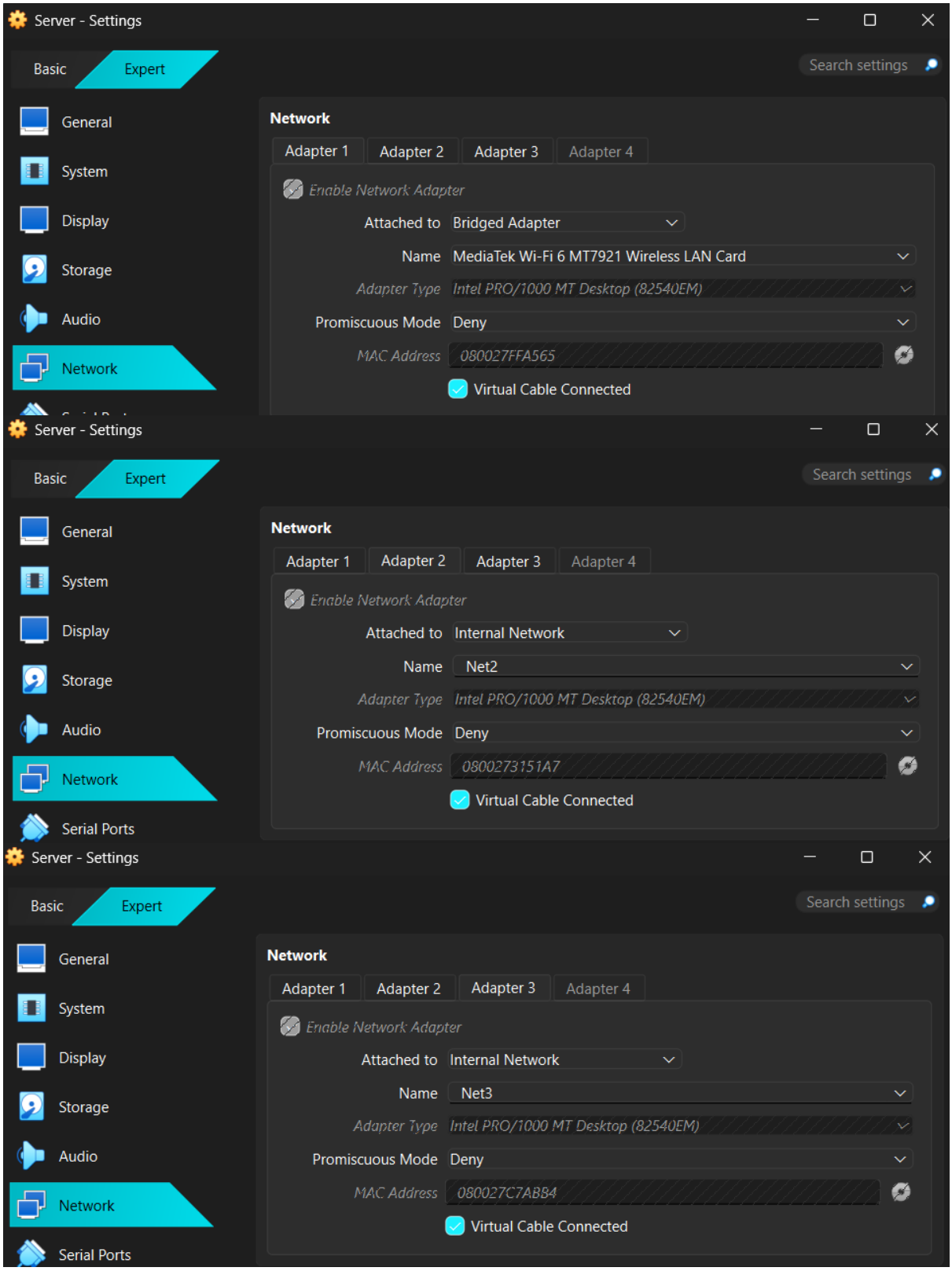
Computer Networks: Course Project

Author
Anvarbek Makhmudov

Stage 1

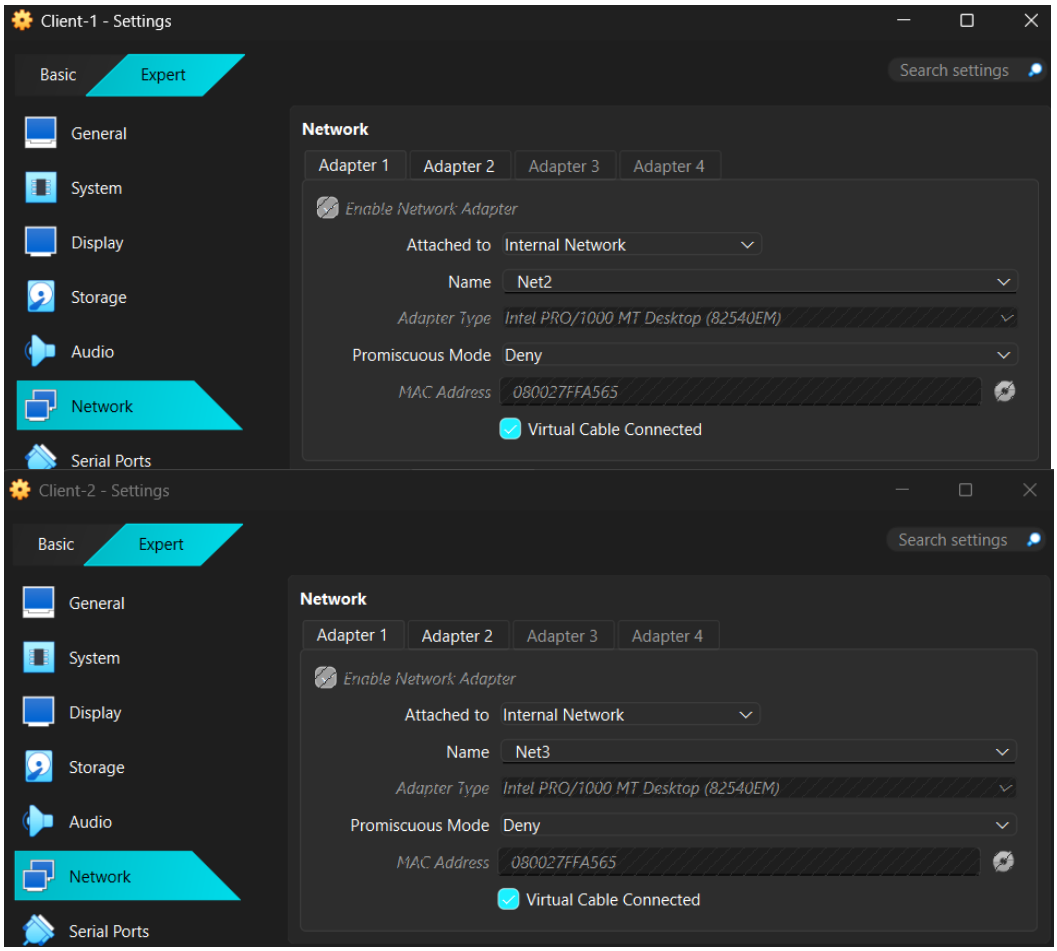
Creating and Configuring a Virtual Machine

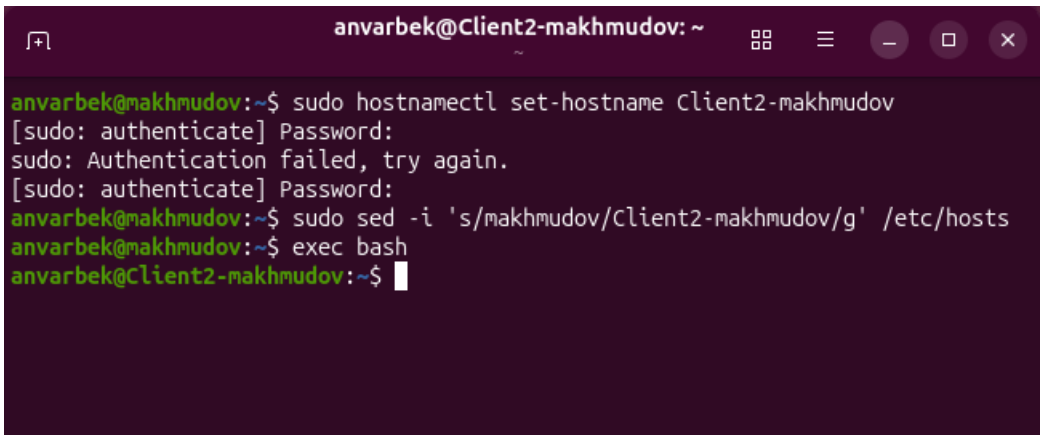


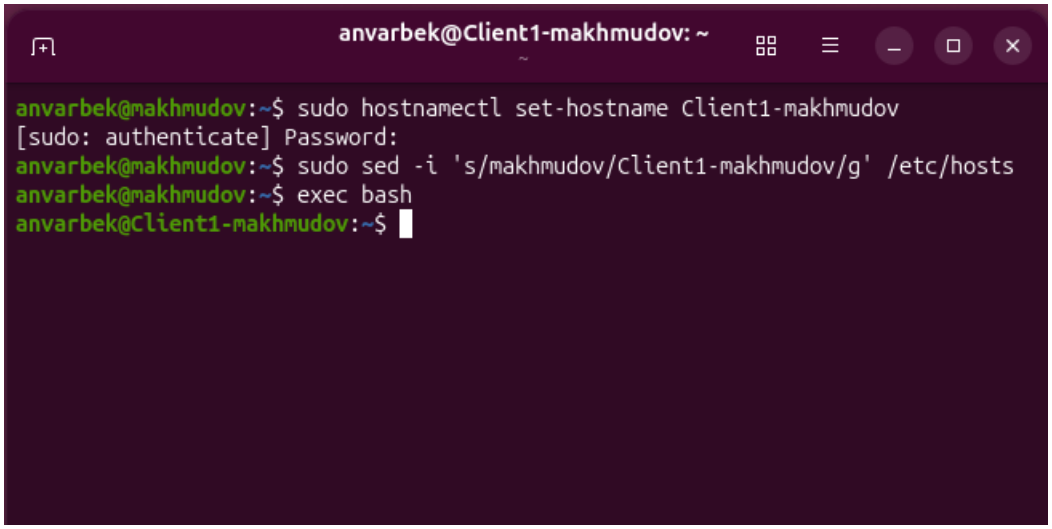

Server - Settings
Basic
Expert
Search settings
General
System
Display
Storage
Audio
Network
Network
Adapter 1
Adapter 2
Adapter 3
Adapter 4
Enable Network Adapter
Attached to
Bridged Adapter
Name
MediaTek Wi-Fi 6 MT7921 Wireless LAN Card
Adapter Type
Intel PRO/1000 MT Desktop (82540EM)
Promiscuous Mode
Deny
MAC Address
080027FFA565
Virtual Cable Connected
Server - Settings
Basic
Expert
Search settings
General
System
Display
Storage
Audio
Network
Serial Ports
Network
Adapter 1
Adapter 2
Adapter 3
Adapter 4
Enable Network Adapter
Attached to
Internal Network
Name
Net2
Adapter Type
Intel PRO/1000 MT Desktop (82540EM)
Promiscuous Mode
Deny
MAC Address
0800273151A7
Virtual Cable Connected
Server - Settings
Basic
Expert
Search settings
General
System
Display
Storage
Audio
Network
Serial Ports
Network
Adapter 1
Adapter 2
Adapter 3
Adapter 4
Enable Network Adapter
Attached to
Internal Network
Name
Net3
Adapter Type
Intel PRO/1000 MT Desktop (82540EM)
Promiscuous Mode
Deny
MAC Address
080027C7ABB4
Virtual Cable Connected

Username: anvarbek
Server Hostname: makhmudov

Clones: Client1-makhmudov / Client2-makhmudov

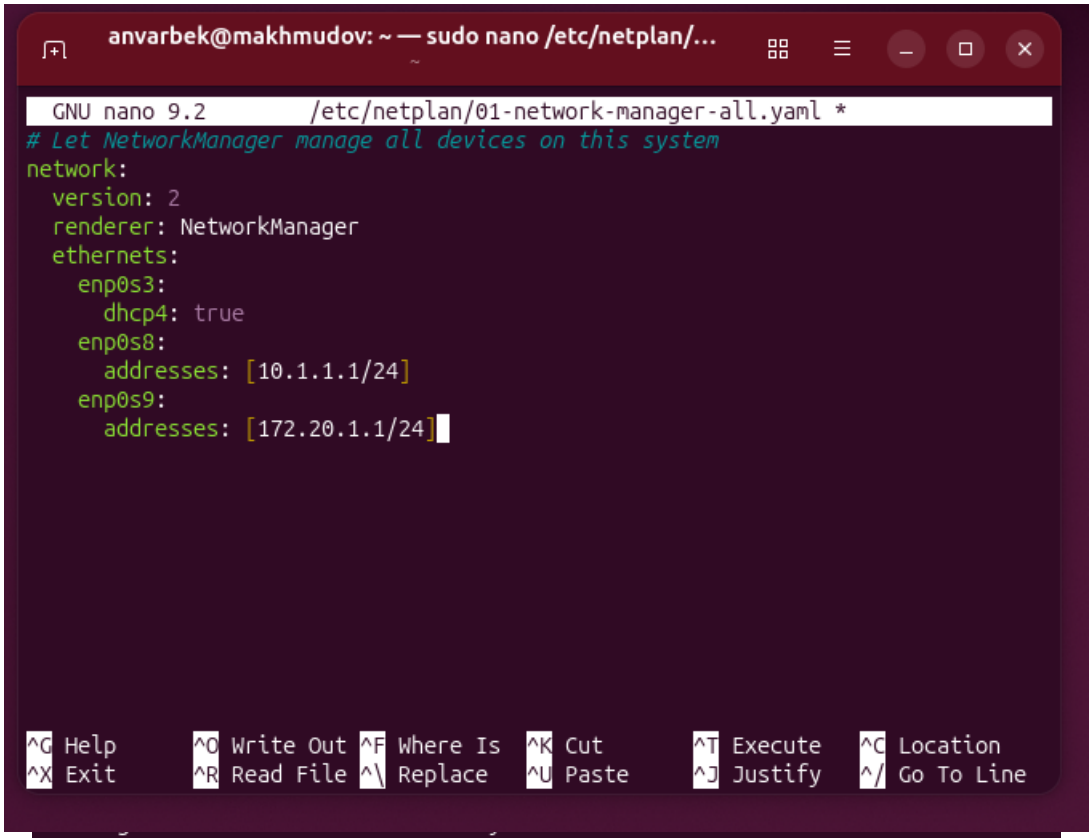






Stage 2

Creating a Network and Configuring the IP Address

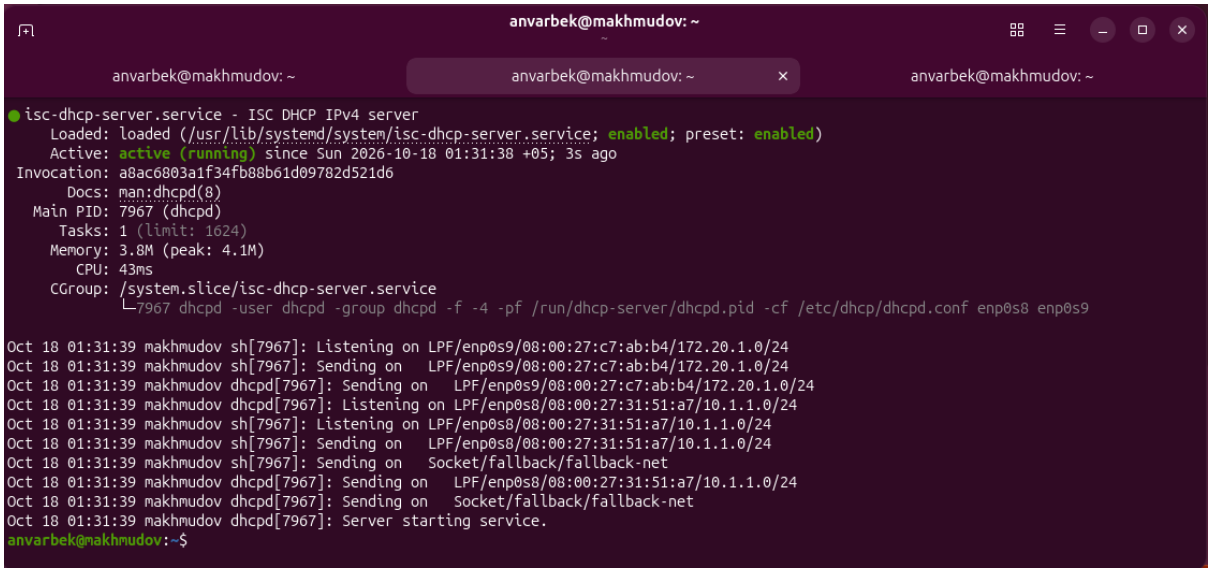



```
anvarbek@makhmudov:~$ ip -br addr
lo               UNKNOWN        127.0.0.1/8 ::1/128
enp0s3           UP             192.168.1.14/24 fe80::a00:27ff:feff:a565/64
enp0s8           UP             10.1.1.1/24 fe80::a00:27ff:fe31:51a7/64
enp0s9           UP             172.20.1.1/24 fe80::a00:27ff:fec7:abb4/64
anvarbek@makhmudov:~$
```

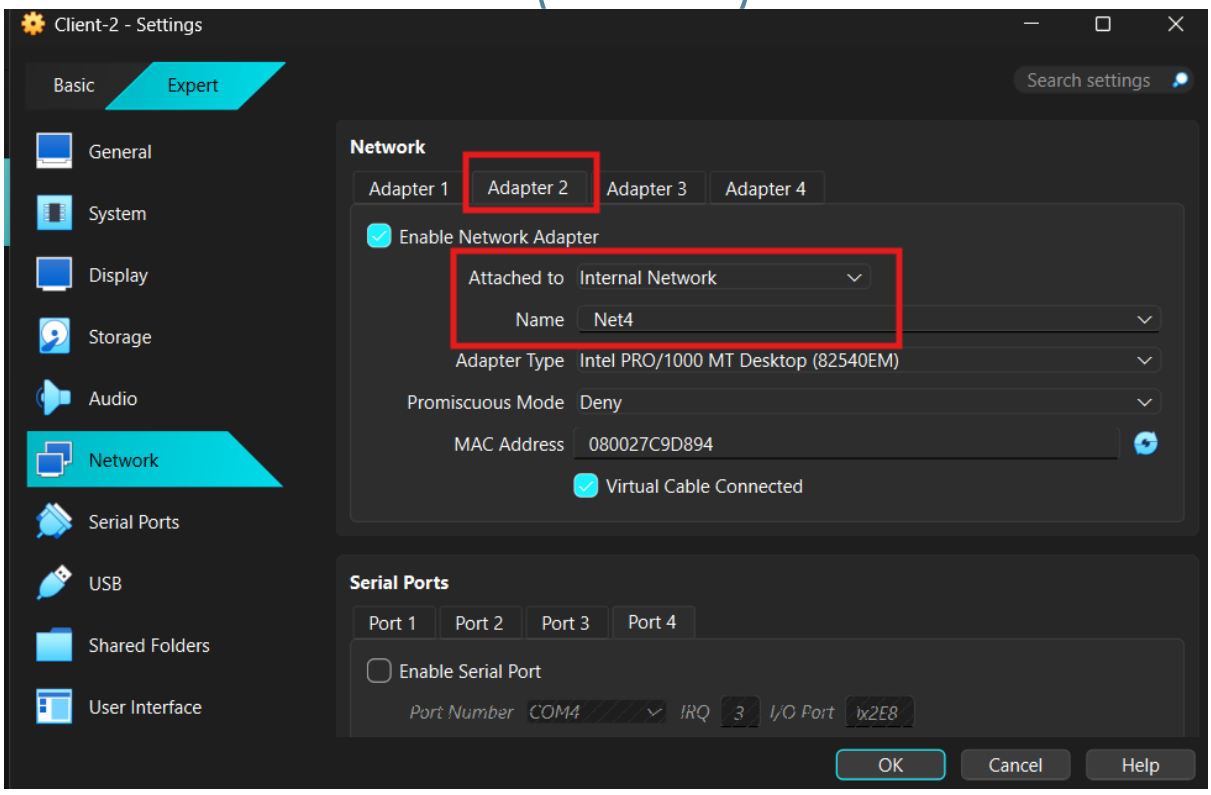
Install DHCP and Configure

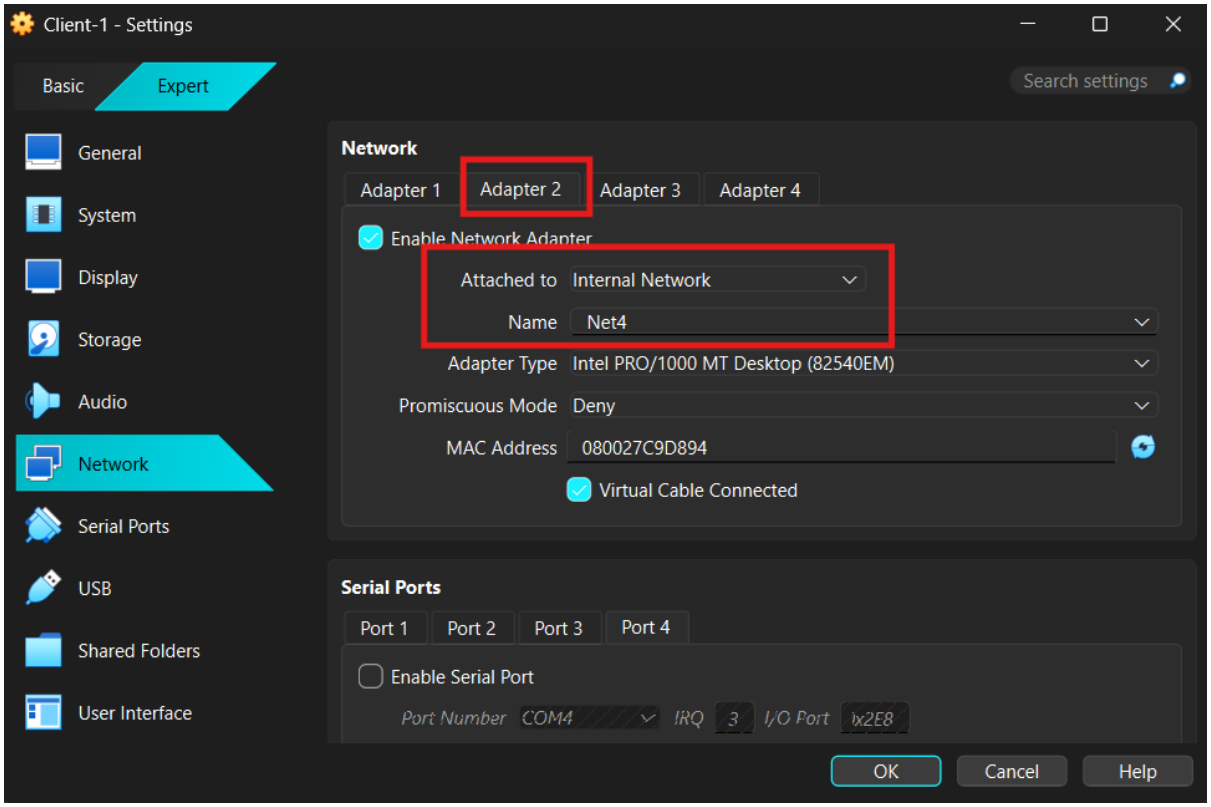


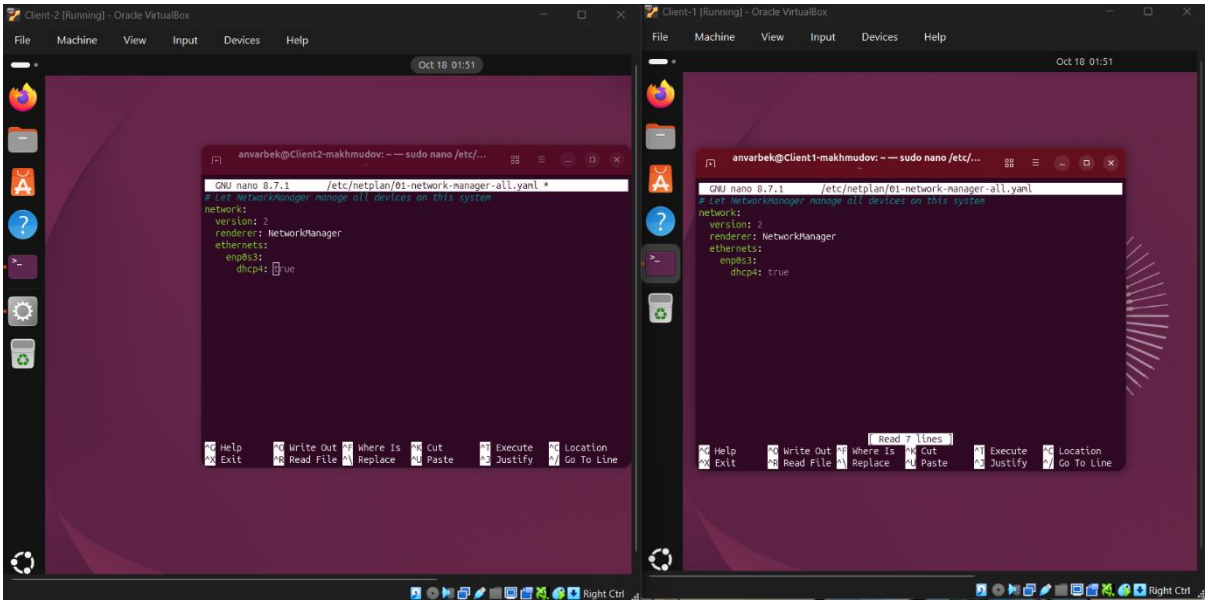
DHCP Status



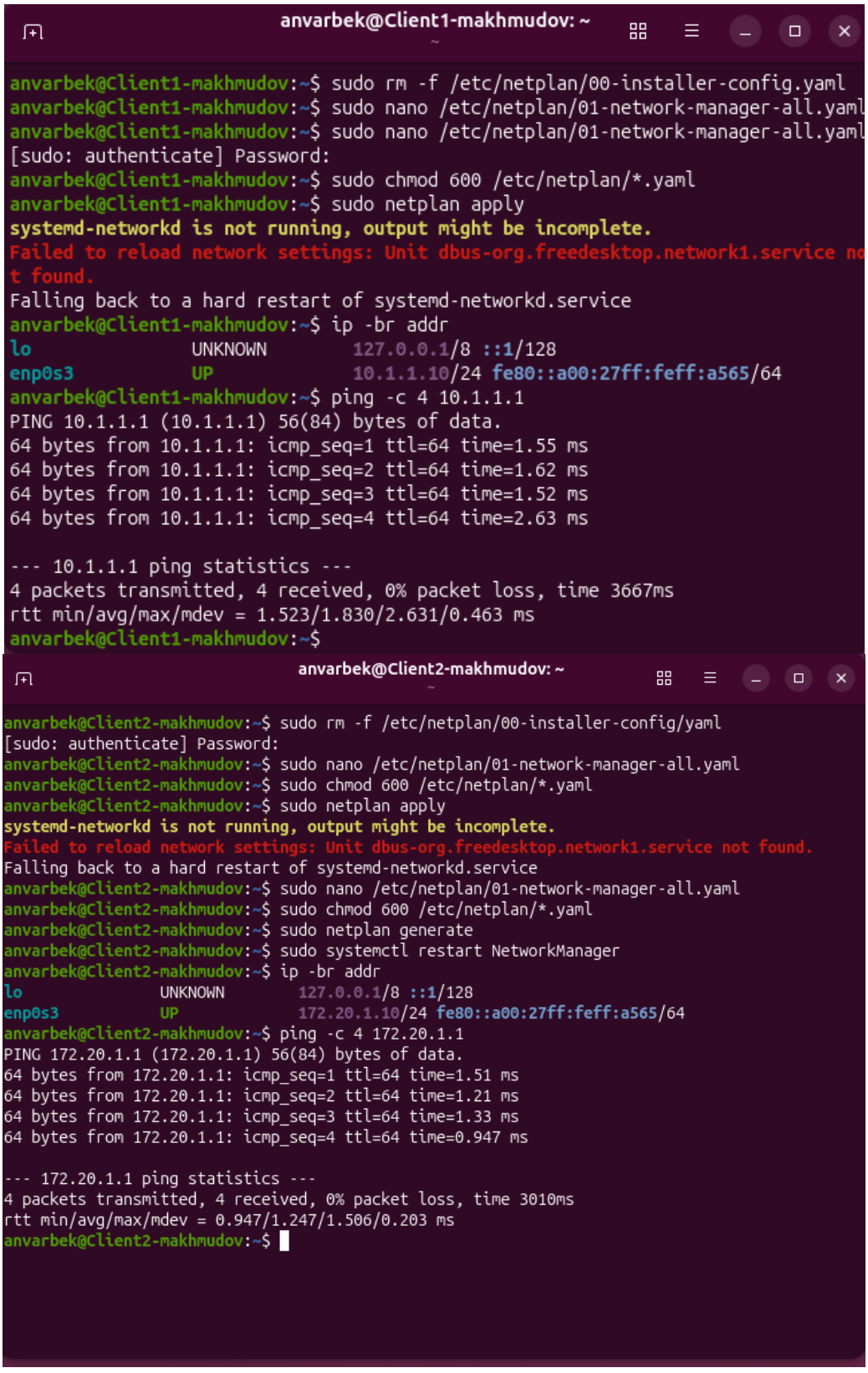
Clients (clones) side




Client-1 - Settings
Basic
Expert
Search settings
General
System
Display
Storage
Audio
Network
Serial Ports
USB
Shared Folders
User Interface
Network
Adapter 1
Adapter 2
Adapter 3
Adapter 4
Enable Network Adapter
Attached to
Internal Network
Name
Net4
Adapter Type
Intel PRO/1000 MT Desktop (82540EM)
Promiscuous Mode
Deny
MAC Address
080027C9D894
Virtual Cable Connected
Serial Ports
Port 1
Port 2
Port 3
Port 4
Enable Serial Port
Port Number
COM4
IRQ
3
I/O Port
0x2E8
OK
Cancel
Help


Client-2 [Running] - Oracle VirtualBox
Client-1 [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
Oct 18 01:51
anvarbek@Client2-makhmudov: ~ — sudo nano /etc/...
anvarbek@Client1-makhmudov: ~ — sudo nano /etc/...
GNU nano 8.7.1
/etc/netplan/01-network-manager-all.yaml
network:
version: 2
renderer: NetworkManager
ethernets:
enp0s3:
dhcp4: true
Read 7 lines
Right Ctrl

Stage 3
Network Health Check



```
anvarbek@Client1-makhmudov:~$ ip addr show enp0s3
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group defau
lt qlen 1000
    link/ether 08:00:27:ff:a5:65 brd ff:ff:ff:ff:ff:ff
    inet 10.1.1.10/24 brd 10.1.1.255 scope global dynamic noprefixroute enp0s3
       valid_lft 336sec preferred_lft 336sec
    inet6 fe80::a00:27ff:feff:a565/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
anvarbek@Client1-makhmudov:~$
```

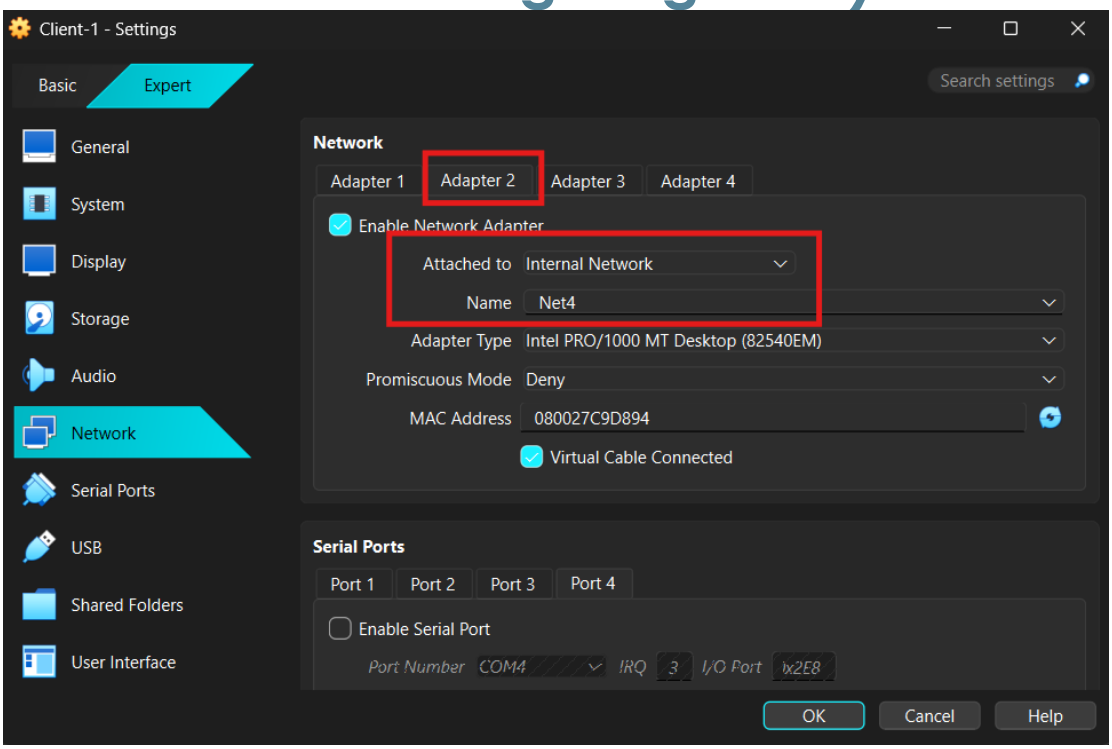
```
anvarbek@Client2-makhmudov:~$ ip addr show enp0s3
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default q
len 1000
    link/ether 08:00:27:ff:a5:65 brd ff:ff:ff:ff:ff:ff
    inet 172.20.1.10/24 brd 172.20.1.255 scope global dynamic noprefixroute enp0s3
       valid_lft 397sec preferred_lft 397sec
    inet6 fe80::a00:27ff:feff:a565/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
anvarbek@Client2-makhmudov:~$
```

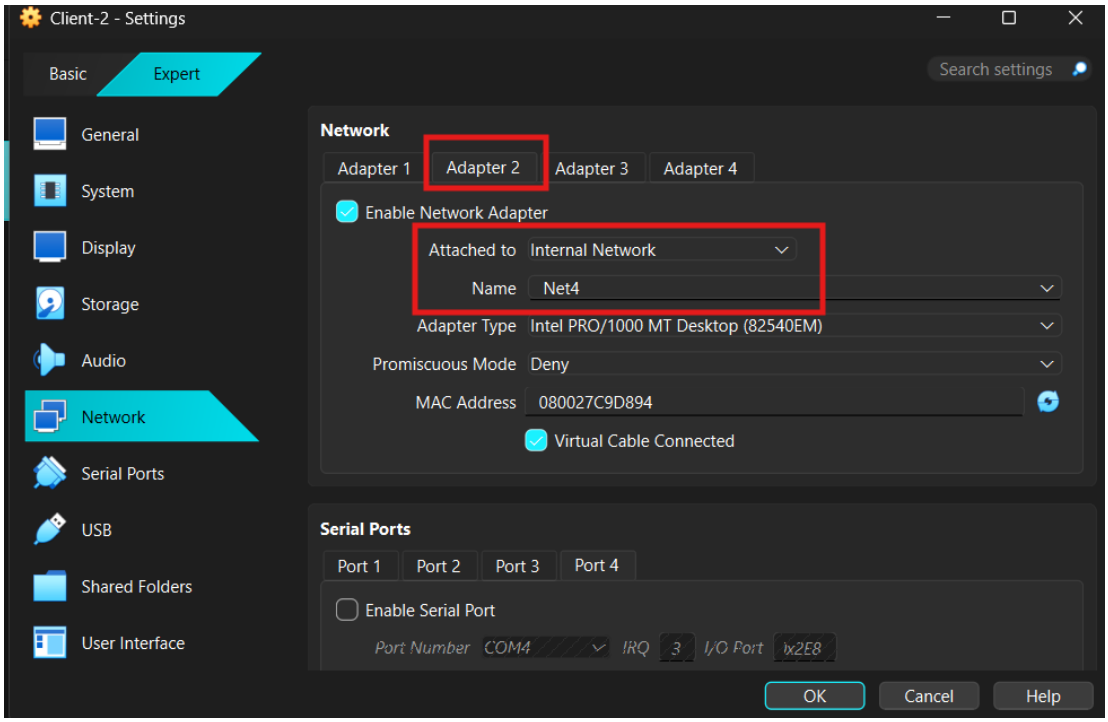
```
anvarbek@makhmudov: ~
anvarbek@makhmudov: ~ | anvarbek@makhmudov: ~ | anvarbek@makhmudov: ~
anvarbek@makhmudov:~$ echo "net.ipv4.ip_forward=1" | sudo tee -a /etc/sysctl.conf
[sudo: authenticate] Password:
net.ipv4.ip_forward=1
anvarbek@makhmudov:~$ sudo sysctl -p
net.ipv4.ip_forward = 1
anvarbek@makhmudov:~$
```

```
anvarbek@Client1-makhmudov: ~
lo               UNKNOWN        127.0.0.1/8 ::1/128
enp0s3           UP             10.1.1.10/24 fe80::a00:27ff:feff:a565/64
anvarbek@Client1-makhmudov:~$ ping -c 4 10.1.1.1
PING 10.1.1.1 (10.1.1.1) 56(84) bytes of data.
64 bytes from 10.1.1.1: icmp_seq=1 ttl=64 time=1.55 ms
64 bytes from 10.1.1.1: icmp_seq=2 ttl=64 time=1.62 ms
64 bytes from 10.1.1.1: icmp_seq=3 ttl=64 time=1.52 ms
64 bytes from 10.1.1.1: icmp_seq=4 ttl=64 time=2.63 ms

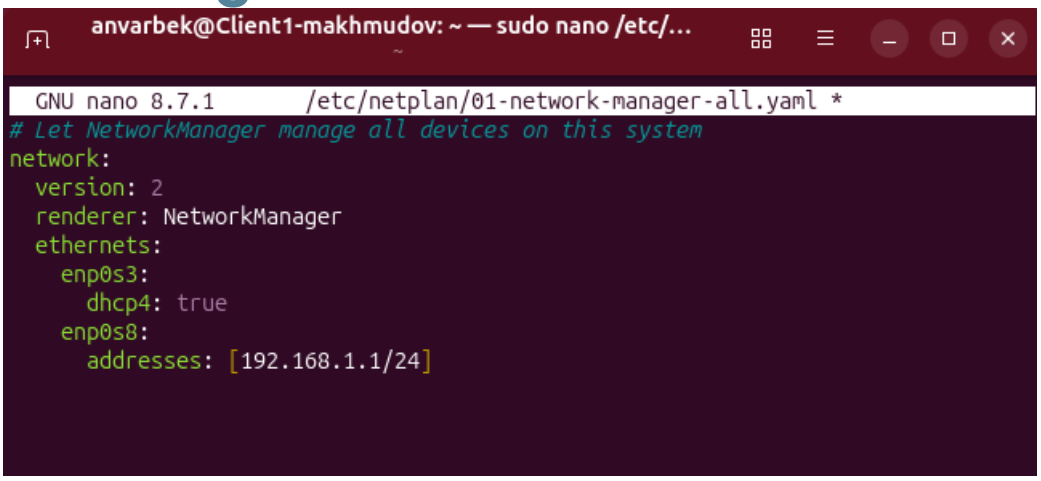
--- 10.1.1.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3667ms
rtt min/avg/max/mdev = 1.523/1.830/2.631/0.463 ms
anvarbek@Client1-makhmudov:~$ ip addr show emp0s3
Device "emp0s3" does not exist.
anvarbek@Client1-makhmudov:~$ sudo netplan apply
anvarbek@Client1-makhmudov:~$ ip addr show emp0s3
Device "emp0s3" does not exist.
anvarbek@Client1-makhmudov:~$ ip addr show enp0s3
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group defau
lt qlen 1000
    link/ether 08:00:27:ff:a5:65 brd ff:ff:ff:ff:ff:ff
    inet 10.1.1.10/24 brd 10.1.1.255 scope global dynamic noprefixroute enp0s3
       valid_lft 336sec preferred_lft 336sec
    inet6 fe80::a00:27ff:feff:a565/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
anvarbek@Client1-makhmudov:~$ traceroute 172.20.1.10
traceroute to 172.20.1.10 (172.20.1.10), 30 hops max, 60 byte packets
 1  10.1.1.1 (10.1.1.1)  1.146 ms  1.165 ms  0.903 ms
 2  172.20.1.10 (172.20.1.10)  2.259 ms  2.553 ms  3.380 ms
anvarbek@Client1-makhmudov:~$
```

Stage 4
Additional Network Configuration (Adding and Configuring Net4)

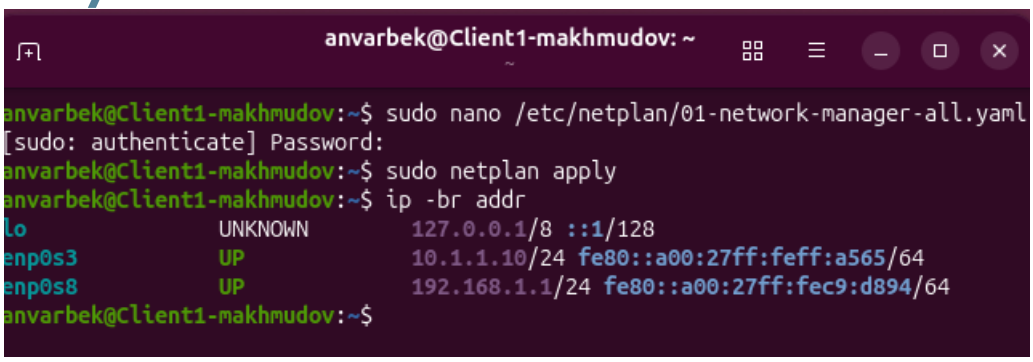




Configure Static Net4 Addresses



Verify Net4 IP Addresses on both Clients





Ping Across Net4

```
anvarbek@Client1-makhmudov:~$ ping -c 4 192.168.1.2
PING 192.168.1.2 (192.168.1.2) 56(84) bytes of data.
64 bytes from 192.168.1.2: icmp_seq=1 ttl=64 time=2.41 ms
64 bytes from 192.168.1.2: icmp_seq=2 ttl=64 time=1.32 ms
64 bytes from 192.168.1.2: icmp_seq=3 ttl=64 time=2.07 ms
64 bytes from 192.168.1.2: icmp_seq=4 ttl=64 time=1.61 ms

--- 192.168.1.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3009ms
rtt min/avg/max/mdev = 1.320/1.854/2.408/0.417 ms
anvarbek@Client1-makhmudov:~$
```

```
anvarbek@Client2-makhmudov: ~

anvarbek@Client2-makhmudov:~$ ip -br addr
lo               UNKNOWN        127.0.0.1/8 ::1/128
enp0s3           UP             172.20.1.10/24 fe80::a00:27ff:feff:a565/64
enp0s8           UP             192.168.1.2/24 fe80::a00:27ff:fec9:cbe0/64
anvarbek@Client2-makhmudov:~$ ping -c 4 192.168.1.1
PING 192.168.1.1 (192.168.1.1) 56(84) bytes of data.
64 bytes from 192.168.1.1: icmp_seq=1 ttl=64 time=0.873 ms
64 bytes from 192.168.1.1: icmp_seq=2 ttl=64 time=0.968 ms
64 bytes from 192.168.1.1: icmp_seq=3 ttl=64 time=1.04 ms
64 bytes from 192.168.1.1: icmp_seq=4 ttl=64 time=0.768 ms

--- 192.168.1.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3177ms
rtt min/avg/max/mdev = 0.768/0.911/1.038/0.101 ms
anvarbek@Client2-makhmudov:~$
```

Stage 5

Configuring and Testing IP Routing

```
anvarbek@Client1-makhmudov:~$ ip route
default via 10.1.1.1 dev enp0s3 proto dhcp src 10.1.1.10 metric 20100
10.1.1.0/24 dev enp0s3 proto kernel scope link src 10.1.1.10 metric 100
192.168.1.0/24 dev enp0s8 proto kernel scope link src 192.168.1.1 metric 101
anvarbek@Client1-makhmudov:~$
```

```
anvarbek@Client2-makhmudov:~$ ip route
default via 172.20.1.1 dev enp0s3 proto dhcp src 172.20.1.10 metric 20100
172.20.1.0/24 dev enp0s3 proto kernel scope link src 172.20.1.10 metric 100
192.168.1.0/24 dev enp0s8 proto kernel scope link src 192.168.1.2 metric 101
anvarbek@Client2-makhmudov:~$
```

Adding direct routes

```
anvarbek@Client1-makhmudov:~$ sudo ip route add 172.20.1.0/24 via 192.168.1.2
anvarbek@Client1-makhmudov:~$ ip route
default via 10.1.1.1 dev enp0s3 proto dhcp src 10.1.1.10 metric 20100
10.1.1.0/24 dev enp0s3 proto kernel scope link src 10.1.1.10 metric 100
172.20.1.0/24 via 192.168.1.2 dev enp0s8
192.168.1.0/24 dev enp0s8 proto kernel scope link src 192.168.1.1 metric 101
anvarbek@Client1-makhmudov:~$
```

```
anvarbek@Client2-makhmudov:~$ sudo ip route add 10.1.1.0/24 via 192.168.1.1
anvarbek@Client2-makhmudov:~$ ip route
default via 172.20.1.1 dev enp0s3 proto dhcp src 172.20.1.10 metric 20100
10.1.1.0/24 via 192.168.1.1 dev enp0s8
172.20.1.0/24 dev enp0s3 proto kernel scope link src 172.20.1.10 metric 100
192.168.1.0/24 dev enp0s8 proto kernel scope link src 192.168.1.2 metric 101
anvarbek@Client2-makhmudov:~$
```

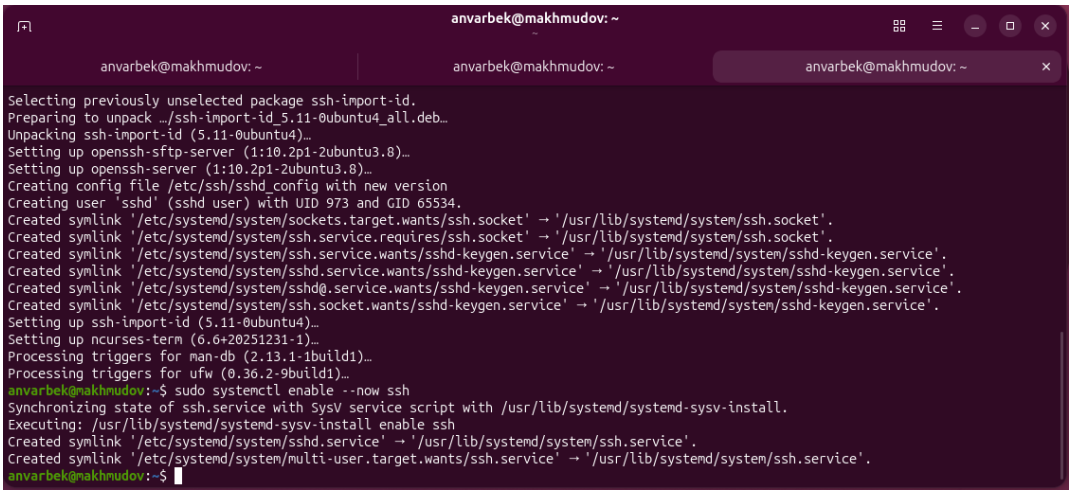
Second Traceroute

```
anvarbek@Client1-makhmudov:~$ traceroute 172.20.1.10
traceroute to 172.20.1.10 (172.20.1.10), 30 hops max, 60 byte packets
 1  172.20.1.10 (172.20.1.10)  1.104 ms  1.064 ms  0.893 ms
anvarbek@Client1-makhmudov:~$
```

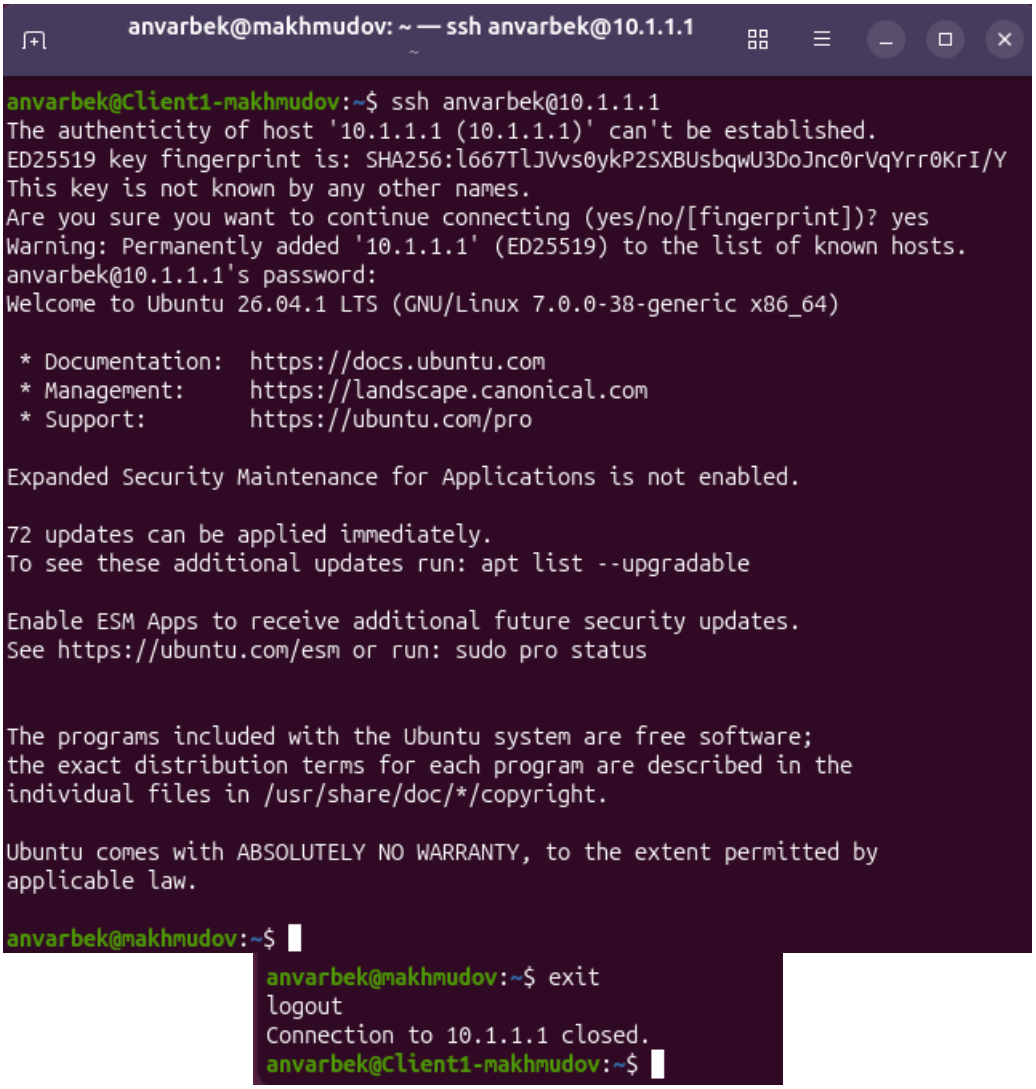
Stage 6

Configuring and Testing SSH

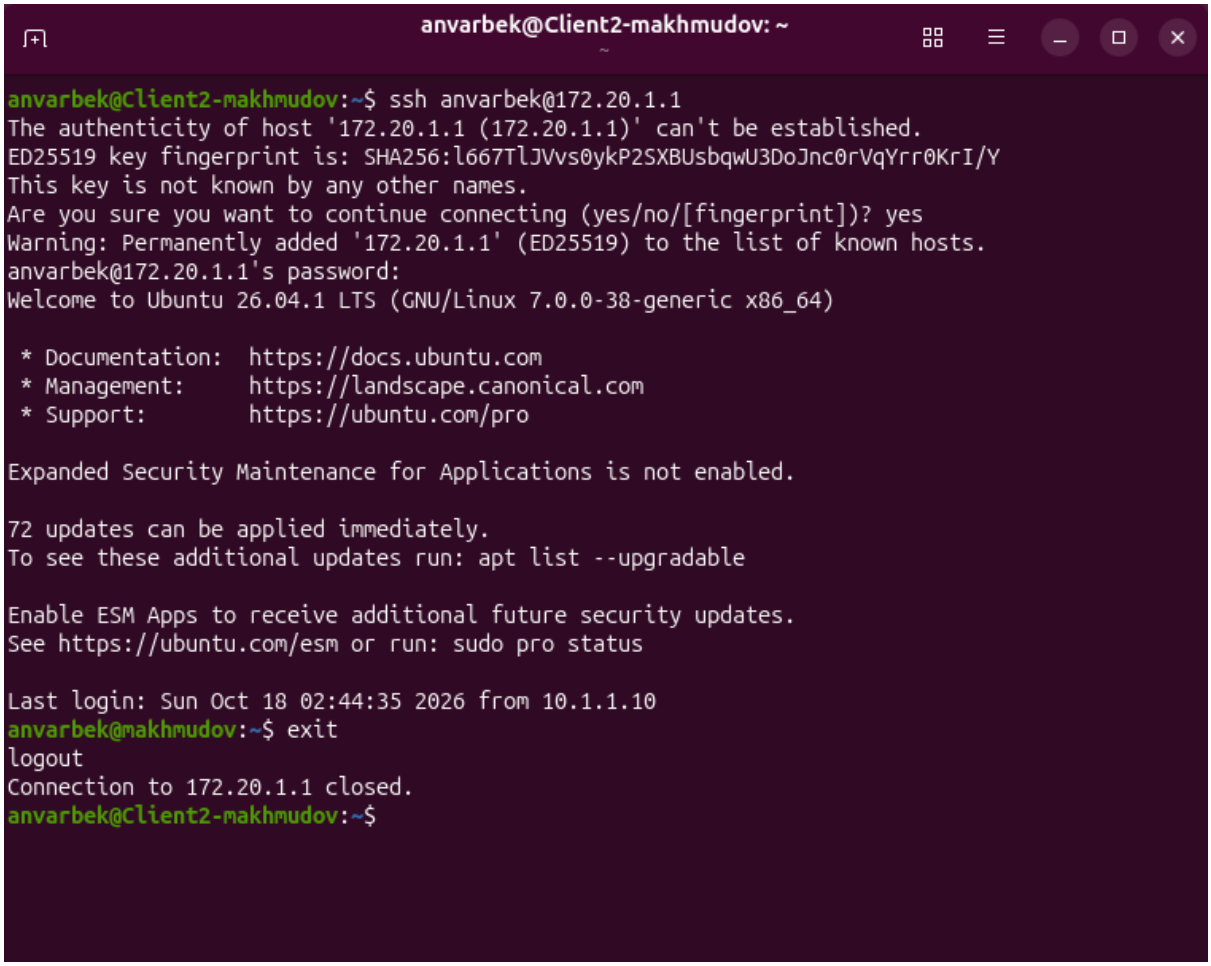
Install



Client1 connect

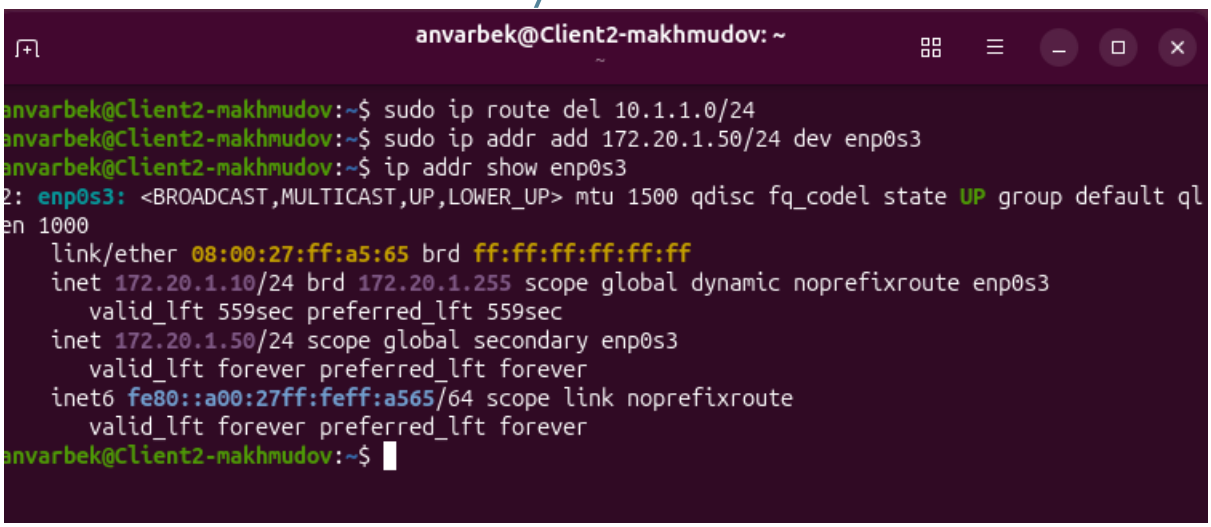


Client2 connect

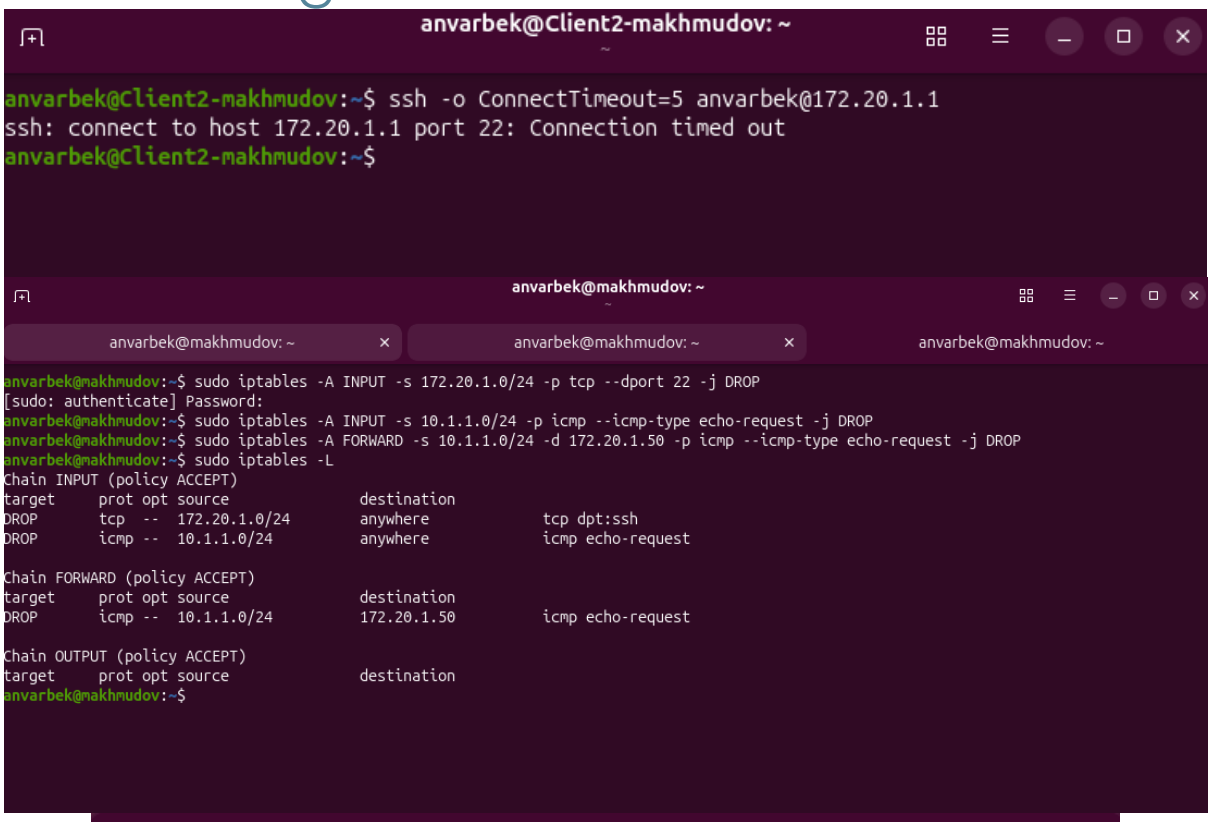


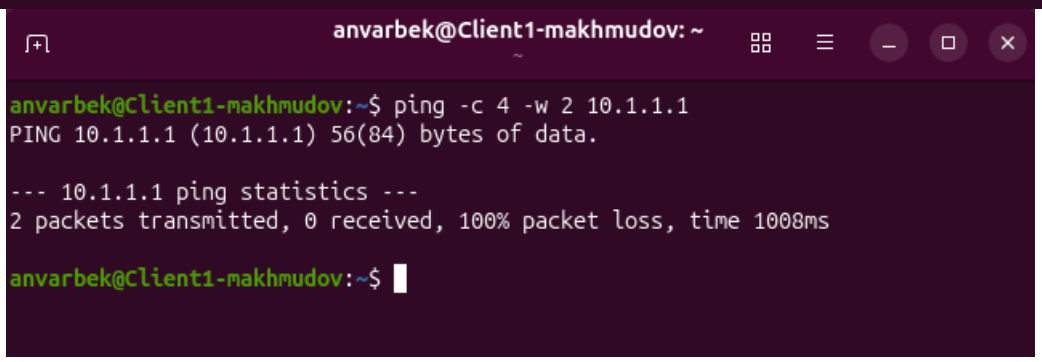
Stage 7
Configuring and Testing Iptables

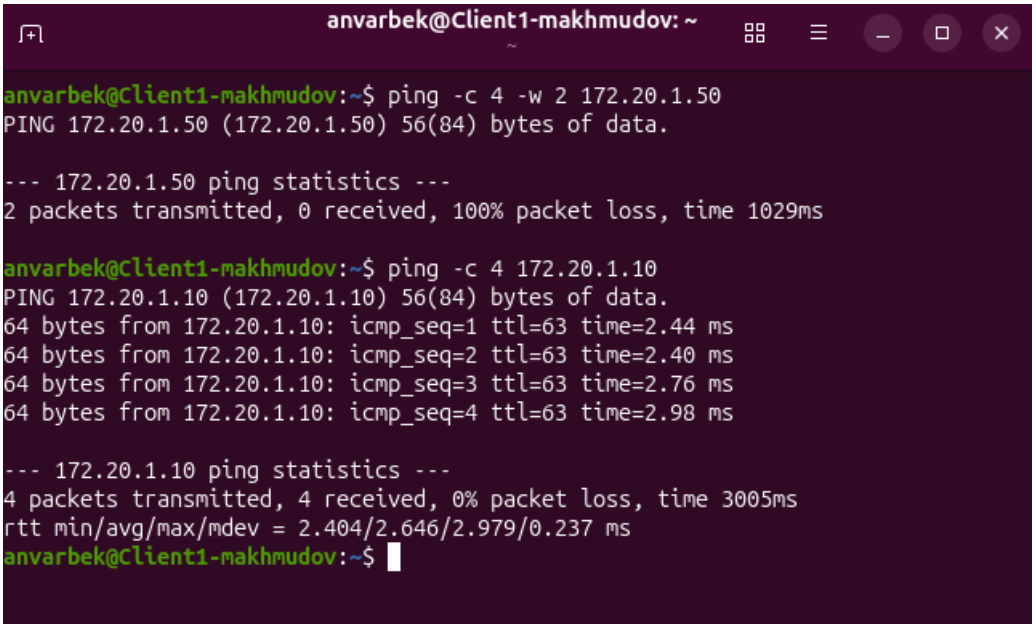
Add Secondary Address on Client2



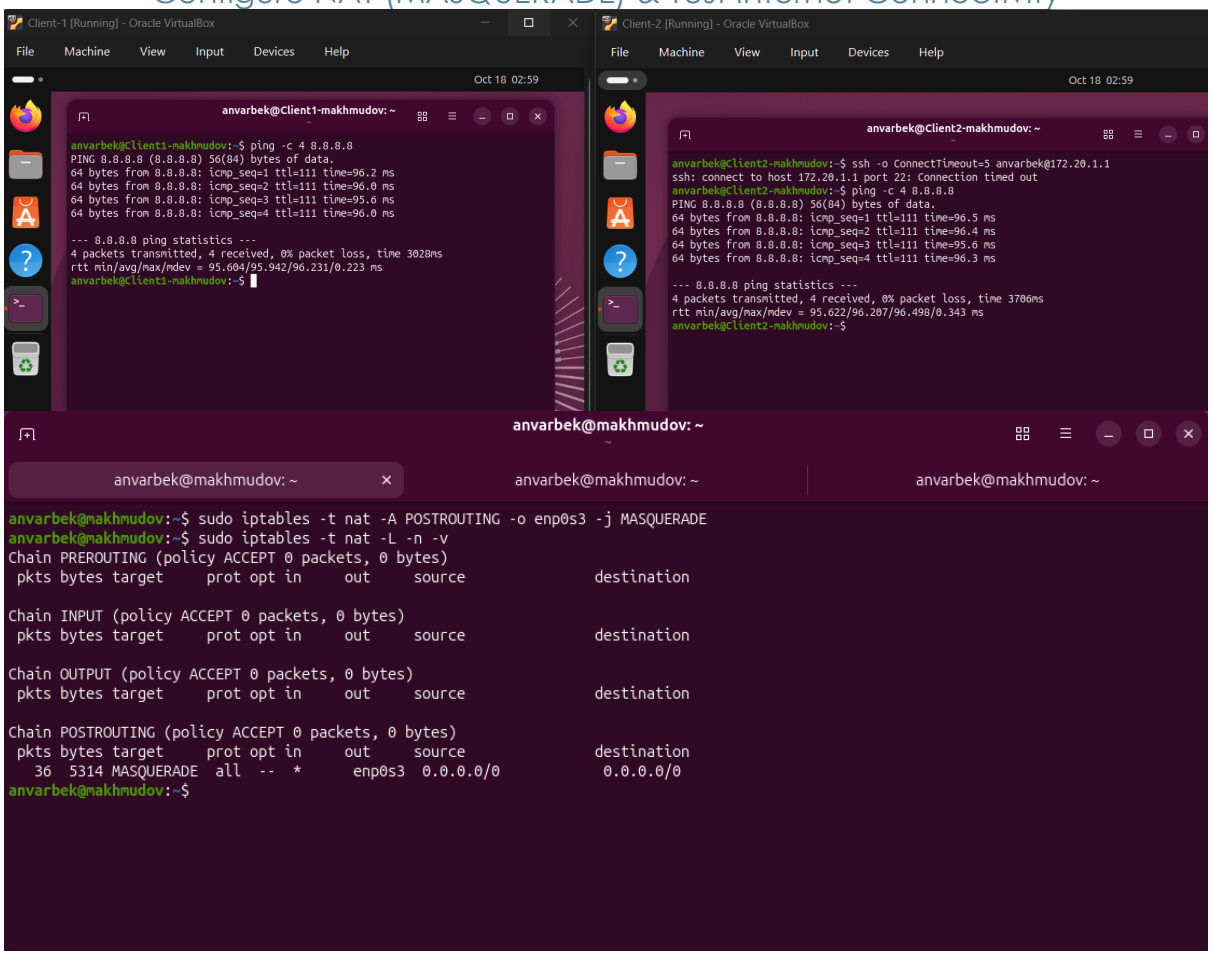
Configure and Test Firewall Rules







Configure NAT (MASQUERADE) & Test Internet Connectivity



THE END